ZHEHAN ZHANG

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EDUCATION

Tongji University Sep 2023 - Jun 2027 (expected)

B.Sc. in Applied Physics, Guohao College; Strong Foundation Program

GPA: **4.25/5.00** | Average: **87/100** | Overall class rank: **4/25**

Selected coursework: Computational Physics; Artificial Intelligence and Physics; Quantum Mechanics.

RESEARCH INTERESTS

LLM agents and multi-agent systems; memory and reliable execution in long-horizon tasks; human-AI collaboration; multimodal learning.

RESEARCH & INDUSTRY EXPERIENCE

REDnote (Xiaohongshu), Lab1327 Jul 2026 - Present

AI Agent Engineering Intern | Loomind

- Develop task-context management for Loomind, a system coordinating agents across extended software development workflows, including requirement analysis, implementation, and verification.
- Maintain versioned snapshots and handoff records of goals, decisions, completed work, validation results, unresolved issues, and next steps so that agents can resume a shared task after interruptions.
- Contribute to capability selection and task verification by recording selection rationale, actual tool use, and evidence against acceptance criteria.
- Implement cloud/local execution coordination using a versioned runner protocol, task leases, durable result queues, and idempotent reporting to support recovery from execution failures.
- Work on an agent development project involving collaboration with the Qwen team; contribute to the software infrastructure supporting agent execution and evaluation.

RESEARCH PROJECTS

ChartMirror Apr 2026 - Present

Full-stack development | Planned submission to PacificVis 2027

- Develop a full-stack prototype that translates a reference image, target dataset, and natural-language request into a visualization through a multi-stage agent workflow.
- Implement visual decomposition, renderer planning, data-role alignment, style analysis, design-rule checking, rendering, and critic-guided repair as inspectable intermediate stages.
- Integrate Python, Flask, pandas, Pydantic, Altair, Plotly, and WordCloud to connect structured data processing with chart generation and output validation.

The Afterlife of a Chart Jan 2026 - Apr 2026

Research contributor and co-author | Unpublished manuscript

- Organized visualization-lineage data and implemented backend logic for LLM-based multi-agent simulation of framing shifts in online discourse.
- Helped code real-world cases using participant roles, platform context, and framing operations to characterize how visualizations change as they circulate.
- Connected agent goals, narrative differences, and allowed operations to forward, modify, or ignore decisions, supporting propagation tracking and analysis of framing changes.

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Curriculum Vitae

RESEARCH EXPERIENCE (CONTINUED)

AffectGPT-RL Reproduction Mar 2026 - Jun 2026

Research experience with Prof. Zheng Lian, Tongji University

- Reproduced components of AffectGPT-RL for multimodal emotion recognition; studied GRPO-based post-training, reward formulations, and experimental evaluation.
- Examined how reward definitions and generated emotion descriptions interact, including the distinction between optimizing a reward and improving recognition quality.
- Gained experience reading research implementations and analyzing reinforcement-learning experiments for open-vocabulary emotion recognition.

AI4MI: Medical Imaging Communication Apr 2026 - Present

Undergraduate research contributor

- Review literature on patient-facing radiology reports, medical visualization, affective design, and trustworthy AI for communicating medical imaging information.
- Participate in interviews with eight clinicians and synthesize explanation needs and information gaps to identify opportunities for patient-facing AI and visualization systems.

Terahertz Scattering Imaging Dec 2023 - Oct 2025

Project lead | Shanghai municipal undergraduate innovation project

- Reproduced published imaging methods, including successive-interference-cancellation-aided spatial transformation imaging (SSTI), for terahertz propagation and scattering.
- Examined reconstruction errors and edge noise in adaptive SSTI; translated physical assumptions and algorithm descriptions into computational implementations.

SELECTED HONORS & AWARDS

National First Prize, China Undergraduate Mathematical Contest in Modeling (2025)
Silver Medal, University Physics Competition (2024 competition) (2025)
Second Prize, Shanghai University Student Experimental Competition (2025)
National First Prize, National College English Translation Competition (2024)
Second-Class Outstanding Undergraduate Scholarship, Tongji University

TECHNICAL SKILLS

Programming and data: Python, MATLAB, TypeScript, SQL, pandas, Pydantic.

Machine learning: PyTorch; LLM/VLM workflows; GRPO reproduction; model evaluation.

Agent and web systems: LangGraph, FastAPI, Flask, React, SQLModel, MySQL, Git.

Scientific computing and visualization: COMSOL, LaTeX, Altair, Plotly, WordCloud.

English: CET-4: 617; CET-6: 528.